

GUJURI ALEKHA

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PROFESSIONAL SUMMARY

Applied data scientist with 2.5 years of experience carrying machine learning work from a rough problem statement through to production. Builds forecasting, classification and regression models, exposes them through FastAPI services, and stays with them through integration testing, release and the issues that surface afterwards. Recent work has centered on applied ML for smart-city systems: irrigation decisioning, ML-driven alerting with AI-generated summaries, and the platform tooling that makes models easier to train, deploy and reuse. Comfortable working across the seam between data science and software delivery, and currently completing an MCA specializing in Machine Learning and Artificial Intelligence.

TECHNICAL SKILLS

Programming & Data: Python, pandas, NumPy, PySpark, Parquet, JSON, CSV, Neo4j

Machine Learning: scikit-learn, time-series forecasting, binary classification, regression, clustering, anomaly detection, feature engineering, lag features, recursive multi-step inference, synthetic data generation, model evaluation and validation

Generative AI: LLM-based summarization of model alerts, prompt design for operator-facing output

Services & APIs: FastAPI, REST API design, Pydantic request models, microservice integration

ML Engineering & MLOps: CI/CD pipelines, model registry, test execution, code coverage, quality gates

PROFESSIONAL EXPERIENCE

Associate Data Scientist

2024 - Present

Trinity Mobility, Bengaluru, India - joined as an intern and moved into the full-time role

- Designed the model deployment and reuse workflow connecting a model registry with Nexus, giving teams one consistent place to store, version and pull trained models instead of repeating setup work for every deployment.
- Built a reusable pretrained-model module with a CI/CD path that brings model creation and deployment down to a single click, removing manual setup that was previously repeated for each new model.
- Added AI-generated summaries to the machine learning alerts published to the ICCC command center platform, so operators can read in plain language what triggered an alert instead of interpreting raw model output.

KEY PROJECTS

Smart Irrigation - ML Pipeline and Command Center Integration

- Designed a three-stage model pipeline that forecasts soil and weather conditions, decides whether irrigation should run, and then predicts how long it should run for - each stage feeding the next, so one request returns a complete irrigation decision rather than a raw number.
- Integrated the trained models into an existing FastAPI service, tested the inference endpoints, and published irrigation alerts with short AI-generated summaries to the ICCC platform for operator use.
- Tools & technologies used: Python, FastAPI, scikit-learn, pandas, NumPy, pyspark, delta-lake

ML Composer - No-Code Machine Learning Platform

- A no-code platform that lets people with no machine learning background build, train and deploy their own models. Ingestion pipelines connect to SQL databases, big data platforms, APIs and local files; users explore, clean and transform data through a visual interface, then train models using regression, classification and clustering algorithms. Built-in evaluation and validation surface performance metrics and visualizations so a model can be judged before it goes anywhere near production.
- Tools & technologies used: Python, Flask, FastAPI, scikit-learn, PySpark, mlflow, evidently AI, scheduler, mlops

EDUCATION

Master of Computer Applications (MCA), Machine Learning & Artificial Intelligence

Feb 2025 - Feb 2027

Lovely Professional University - expected completion Feb 2027

Bachelor of Science (B.Sc.), Physics

Aug 2018 - Sep 2021

Science Degree College, Kukudakhandi | CGPA: 7.7/10